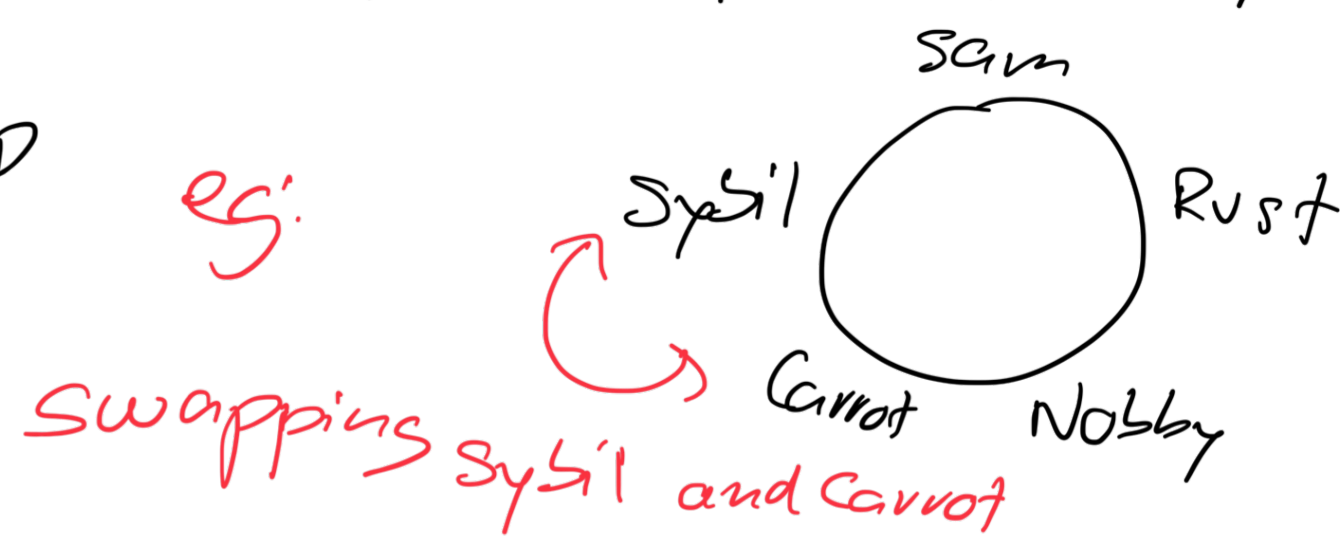


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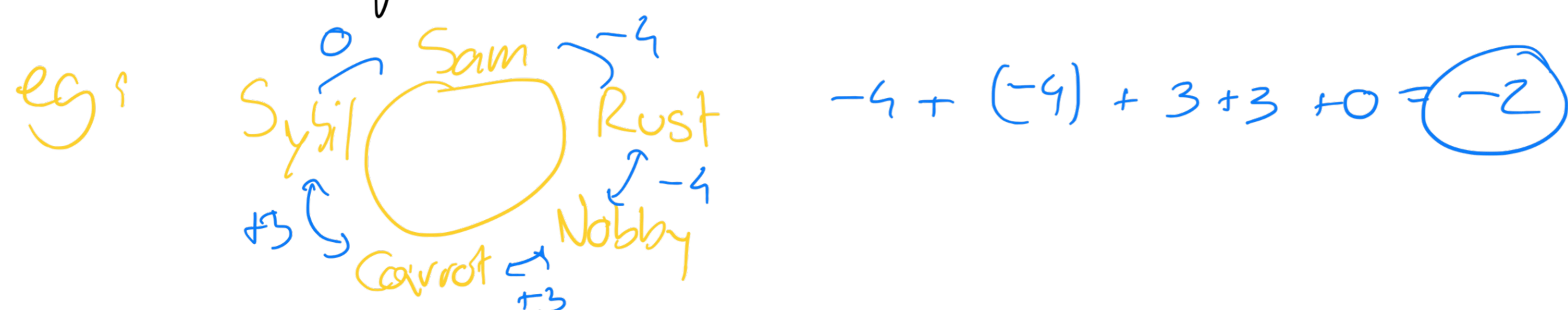
5.1 Nodes represent table configurations eg: [Sam, Rust, Nobby, Carrot, Sybil]

Their neighbors represent a swap of two people in the table

The goal is to find the best total "happiness" value.



The value of a node will be given by the sum of all the values of each pair of guests.



Sy, Sa, Ru, No, Ca -2

Sa	Sy	Ru	No	Ca	3
0	3	-4	2	2	
Ru	Sa	Sy	No	Ca	2
-4	0	4	2	0	
No	Sa	Ru	Sy	Ca	7
2	-4	3	3	3	
Ca	Sa	Ru	No	Sy	-2
2	-4	-4	4	0	
Sy	Ru	Sa	No	Ca	7
3	-4	2	3	3	
Sy	No	Ru	Sa	Ca	1
4	-4	-4	2	3	
Sy	Ca	Ru	No	Sa	1
3	0	-4	2	0	
Sy	Sa	No	Ru	Ca	1
0	2	-4	0	3	
Sy	Sa	Ca	No	Ru	4
0	2	3	-4	3	
Sy	Sa	Ru	Ca	No	3
0	-4	0	3	4	

No, Sa, Ru, Sy, Ca 7

Sa	No	Ru	Sy	Ca	7
2	-4	4	3	2	
Ru	Sa	No	Sy	Ca	5
-4	2	4	3	0	
Sy	Sa	Ru	No	Ca	-6
0	-4	-4	2	0	
Ca	Sa	Ru	Sy	No	8
2	-4	3	4	3	
No	Ru	Sa	Sy	Ca	-3
-4	-4	2	0	3	
No	Sy	Ru	Sa	Ca	8
4	3	-4	2	3	
No	Ca	Ru	Sy	Sa	8
3	0	3	0	2	
No	Sa	Sy	Ru	Ca	4
2	-4	3	0	3	
No	Sa	Ca	Sy	Ru	0
-4	3	3	-4	0	
No	Sa	Ru	Ca	Sy	5
2	-4	0	3	1	

After 2 iterations of the Hill Climbing algorithm,

the best seating order is [Carrot, Sam, Rust, Sybil, Nobby]